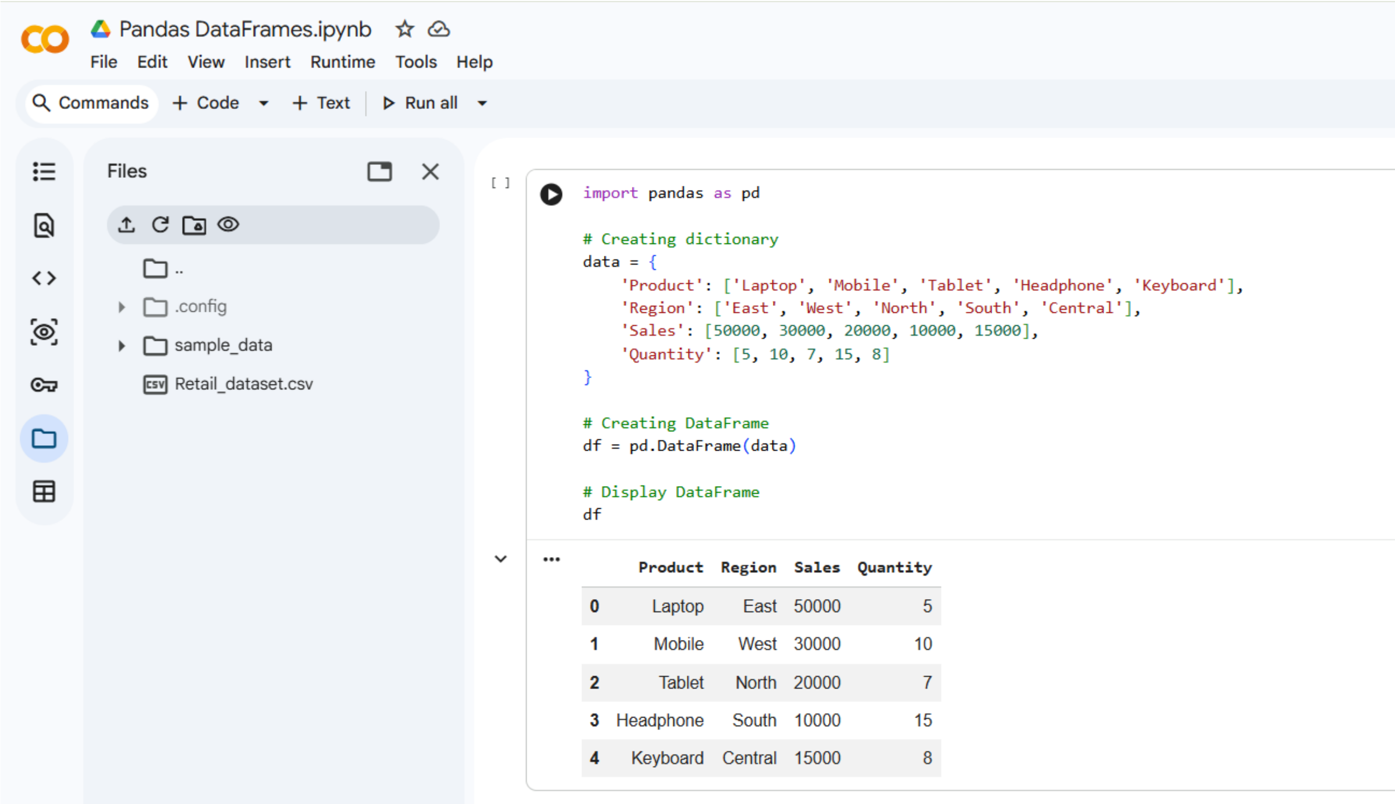
Retail_Dataset

Q1. Creating a DataFrame

Create a Pandas DataFrame using a Python dictionary with the following columns:

- Product
- Region
- Sales
- Quantity

Ans:-




```python
import pandas as pd

# Creating dictionary
data = {
    'Product': ['Laptop', 'Mobile', 'Tablet', 'Headphone', 'Keyboard'],
    'Region': ['East', 'West', 'North', 'South', 'Central'],
    'Sales': [50000, 30000, 20000, 10000, 15000],
    'Quantity': [5, 10, 7, 15, 8]
}

# Creating DataFrame
df = pd.DataFrame(data)

# Display DataFrame
df
```

	Product	Region	Sales	Quantity
0	Laptop	East	50000	5
1	Mobile	West	30000	10
2	Tablet	North	20000	7
3	Headphone	South	10000	15
4	Keyboard	Central	15000	8

Q2. Data Inspection

Perform the following inspection tasks:

1. Display the first 5 rows of the dataset.
2. Display the last 5 rows.
3. Show the structure of the dataset using info().
4. Display the shape of the dataset.

Ans:-

Pandas DataFrames.ipynb

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sample_data
Retail_dataset.csv

```python
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Display the first 5 rows
print("First 5 Rows:")
display(df.head())

# 2. Display the last 5 rows
print("Last 5 Rows:")
display(df.tail())

# 3. Show dataset structure
print("Dataset Information:")
df.info()

# 4. Display dataset shape
print("\nDataset Shape:")
print(df.shape)
```

First 5 Rows:

	Order_ID	Customer	Product	Region	Sales	Quantity
0	1001	Sneha	Laptop	West	1056	2
1	1002	Neha	Tablet	South	904	7
2	1003	Meera	Laptop	South	1597	5
3	1004	Rohan	Keyboard	North	777	8
4	1005	Priya	Phone	South	2028	4

Last 5 Rows:

	Order_ID	Customer	Product	Region	Sales	Quantity
85	1086	Pooja	Laptop	East	1330	2
86	1087	Anita	Keyboard	West	741	3
87	1088	Karan	Monitor	East	875	2
88	1089	Raj	Tablet	North	2213	5
89	1090	Priya	Laptop	South	1270	2

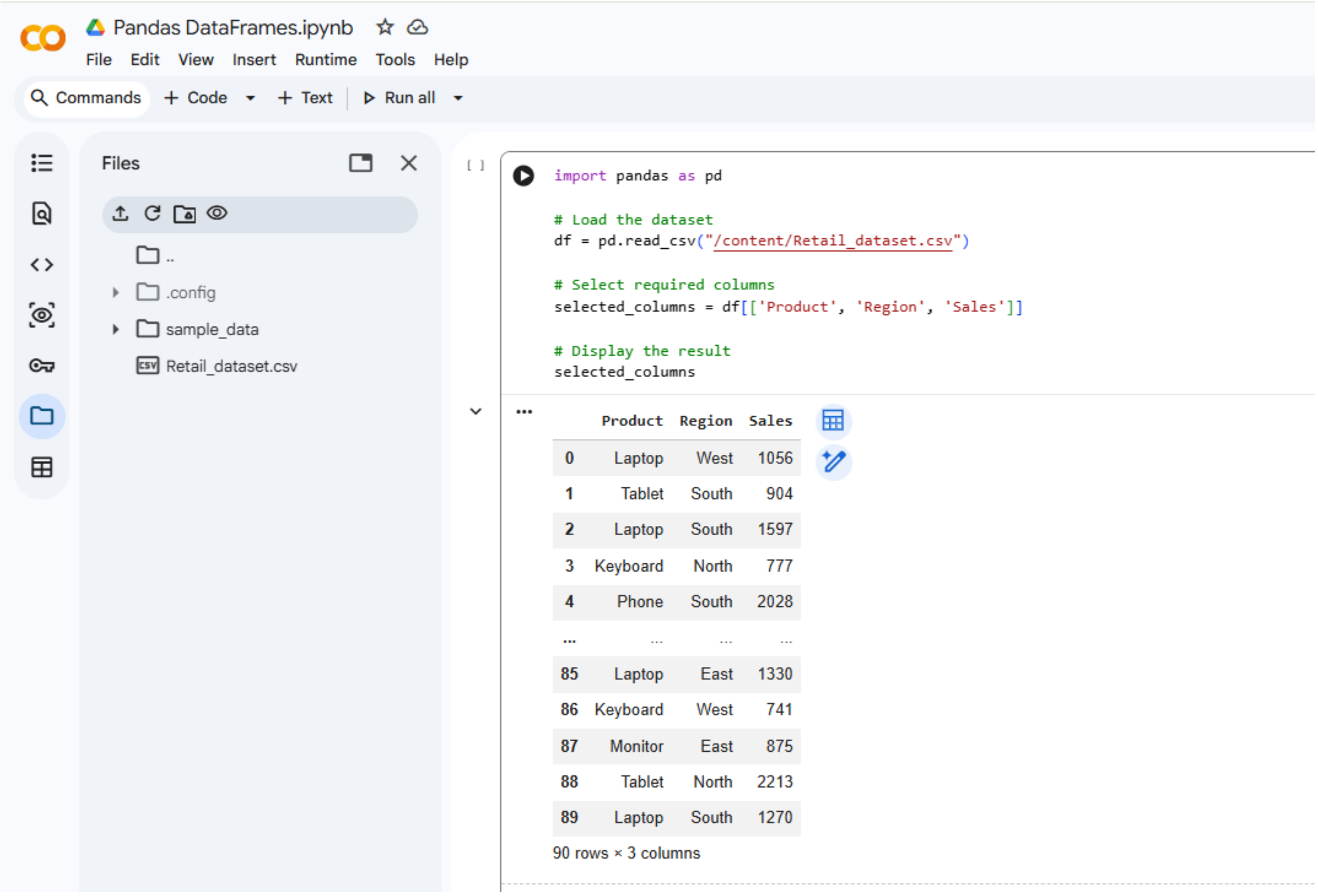
```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 90 entries, 0 to 89
Data columns (total 6 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Order_ID  90 non-null     int64
 1   Customer  90 non-null     object
 2   Product   90 non-null     object
 3   Region    90 non-null     object
 4   Sales     90 non-null     int64
 5   Quantity  90 non-null     int64
dtypes: int64(3), object(3)
memory usage: 4.3+ KB

Dataset Shape:
(90, 6)
```

Q3. Column Selection

Select and display only the following columns:

- Product
- Region
- Sales

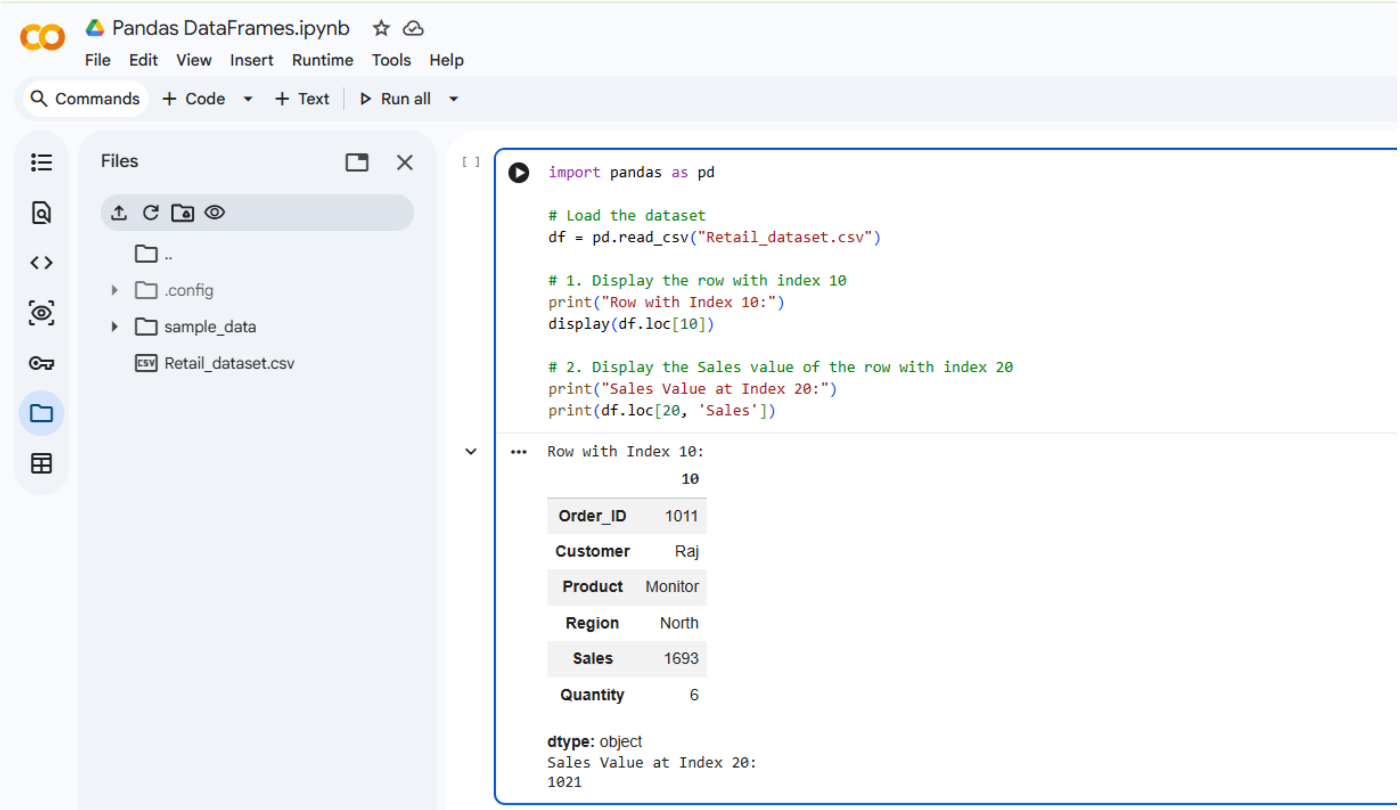
Ans:-



Q4. Indexing using .loc, perform the following:

1. Display the row with index 10.
2. Display the Sales value of the row with index 20

Ans:-




```python
import pandas as pd

# Load the dataset
df = pd.read_csv("Retail_dataset.csv")

# 1. Display the row with index 10
print("Row with Index 10:")
display(df.loc[10])

# 2. Display the Sales value of the row with index 20
print("Sales Value at Index 20:")
print(df.loc[20, 'Sales'])
```

Row with Index 10:

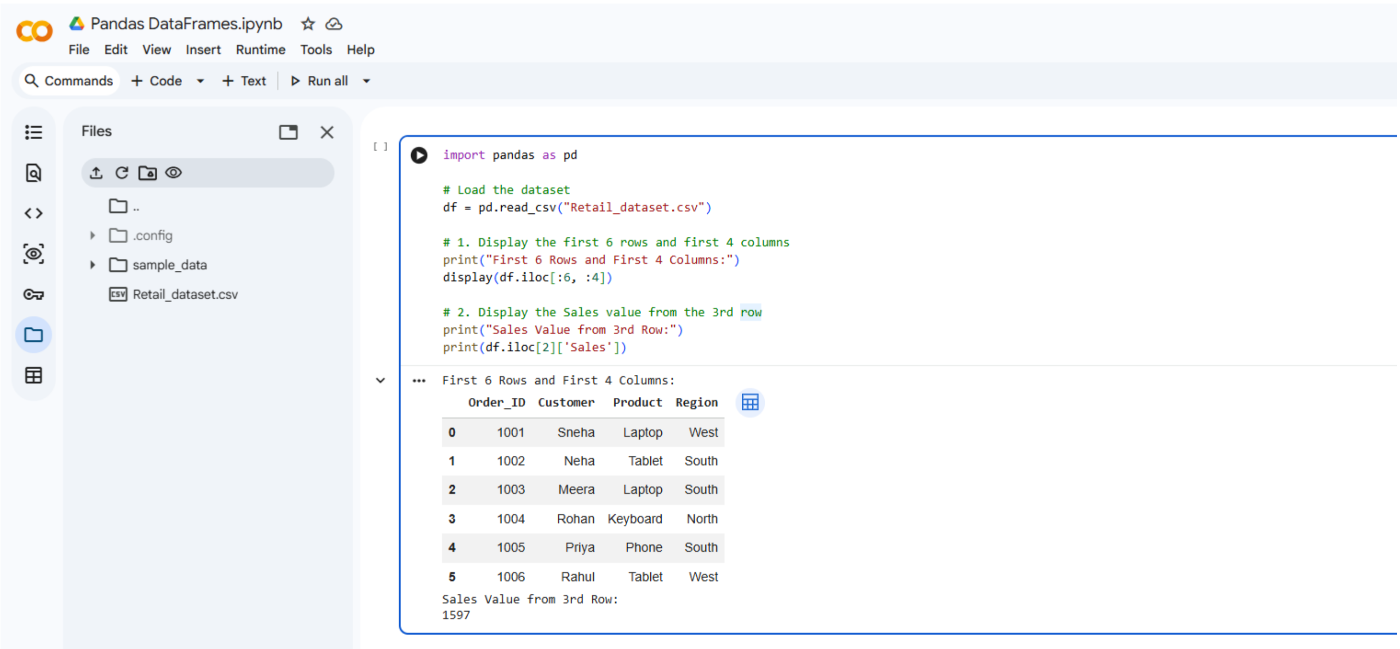
	10
Order_ID	1011
Customer	Raj
Product	Monitor
Region	North
Sales	1693
Quantity	6

dtype: object
Sales Value at Index 20:
1021

Q5. Indexing using .iloc
Using .iloc, display:

1. The first 6 rows and first 4 columns.
2. The Sales value from the 3rd row.

Ans:-




```python
import pandas as pd

# Load the dataset
df = pd.read_csv("Retail_dataset.csv")

# 1. Display the first 6 rows and first 4 columns
print("First 6 Rows and First 4 Columns:")
display(df.iloc[:6, :4])

# 2. Display the Sales value from the 3rd row
print("Sales Value from 3rd Row:")
print(df.iloc[2]['Sales'])
```

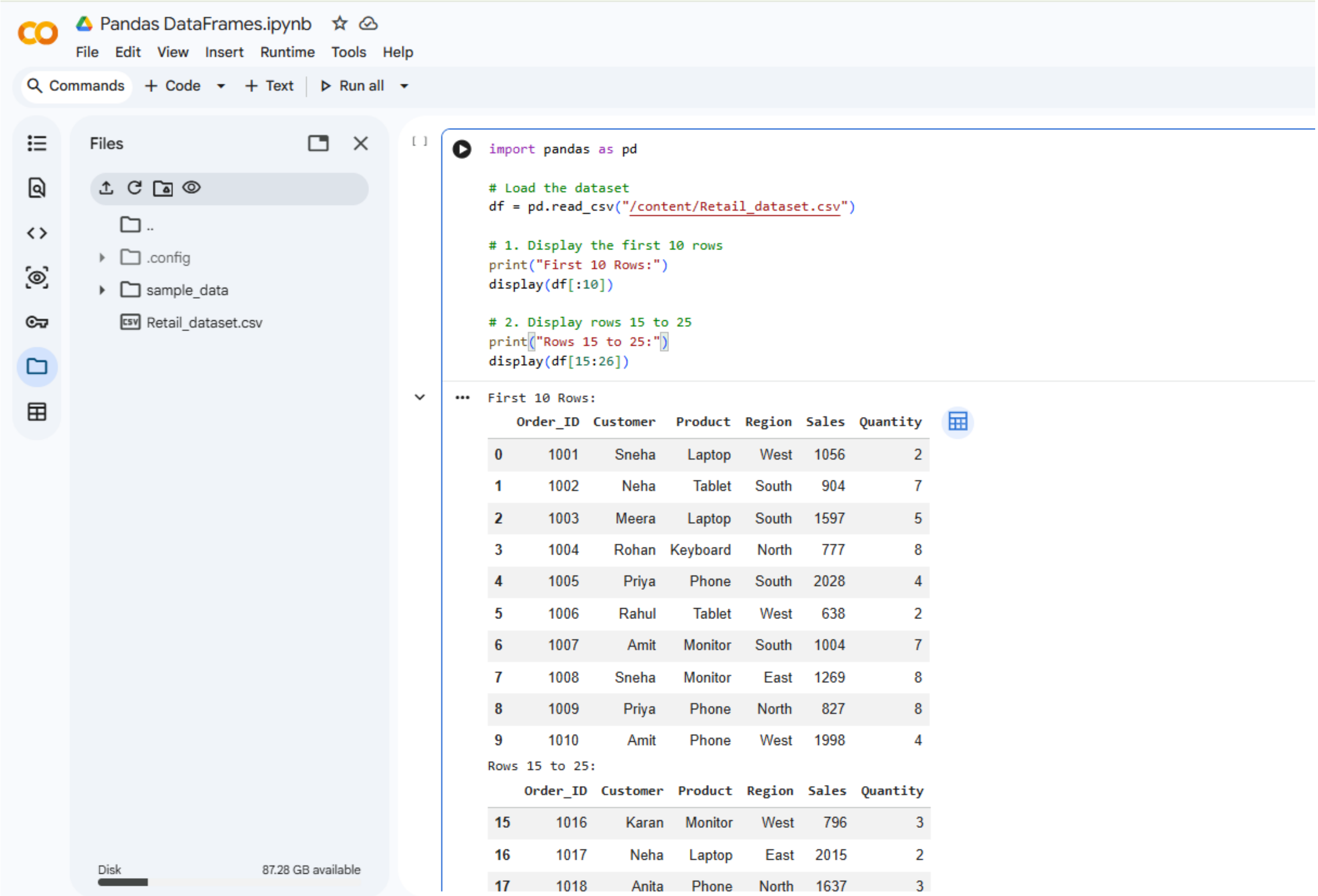
First 6 Rows and First 4 Columns:

	Order_ID	Customer	Product	Region
0	1001	Sneha	Laptop	West
1	1002	Neha	Tablet	South
2	1003	Meera	Laptop	South
3	1004	Rohan	Keyboard	North
4	1005	Priya	Phone	South
5	1006	Rahul	Tablet	West

Sales Value from 3rd Row:
1597

Q6. Row Slicing
Display:

1. The first 10 rows of the dataset.
2. Rows 15 to 25.



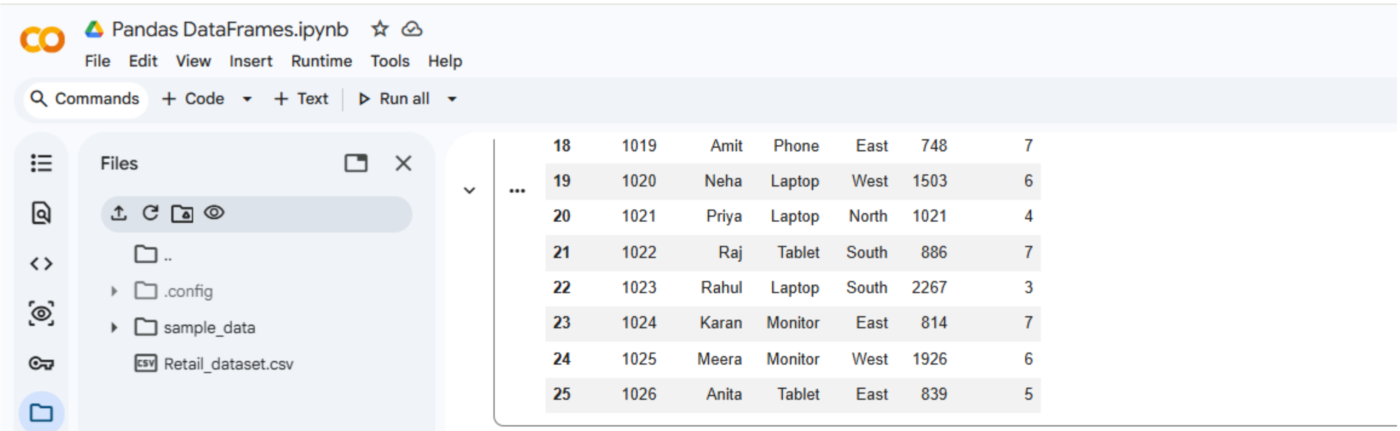

```python
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Display the first 10 rows
print("First 10 Rows:")
display(df[:10])

# 2. Display rows 15 to 25
print("Rows 15 to 25:")
display(df[15:26])
```

First 10 Rows:

	Order_ID	Customer	Product	Region	Sales	Quantity
0	1001	Sneha	Laptop	West	1056	2
1	1002	Neha	Tablet	South	904	7
2	1003	Meera	Laptop	South	1597	5
3	1004	Rohan	Keyboard	North	777	8
4	1005	Priya	Phone	South	2028	4
5	1006	Rahul	Tablet	West	638	2
6	1007	Amit	Monitor	South	1004	7
7	1008	Sneha	Monitor	East	1269	8
8	1009	Priya	Phone	North	827	8
9	1010	Amit	Phone	West	1998	4

Rows 15 to 25:

	Order_ID	Customer	Product	Region	Sales	Quantity
15	1016	Karan	Monitor	West	796	3
16	1017	Neha	Laptop	East	2015	2
17	1018	Anita	Phone	North	1637	3



18	1019	Amit	Phone	East	748	7
19	1020	Neha	Laptop	West	1503	6
20	1021	Priya	Laptop	North	1021	4
21	1022	Raj	Tablet	South	886	7
22	1023	Rahul	Laptop	South	2267	3
23	1024	Karan	Monitor	East	814	7
24	1025	Meera	Monitor	West	1926	6
25	1026	Anita	Tablet	East	839	5

Q7. Filtering Data

Filter the dataset to display:

1. Rows where Region = "East".
2. Rows where Sales > 1500.

Ans:-

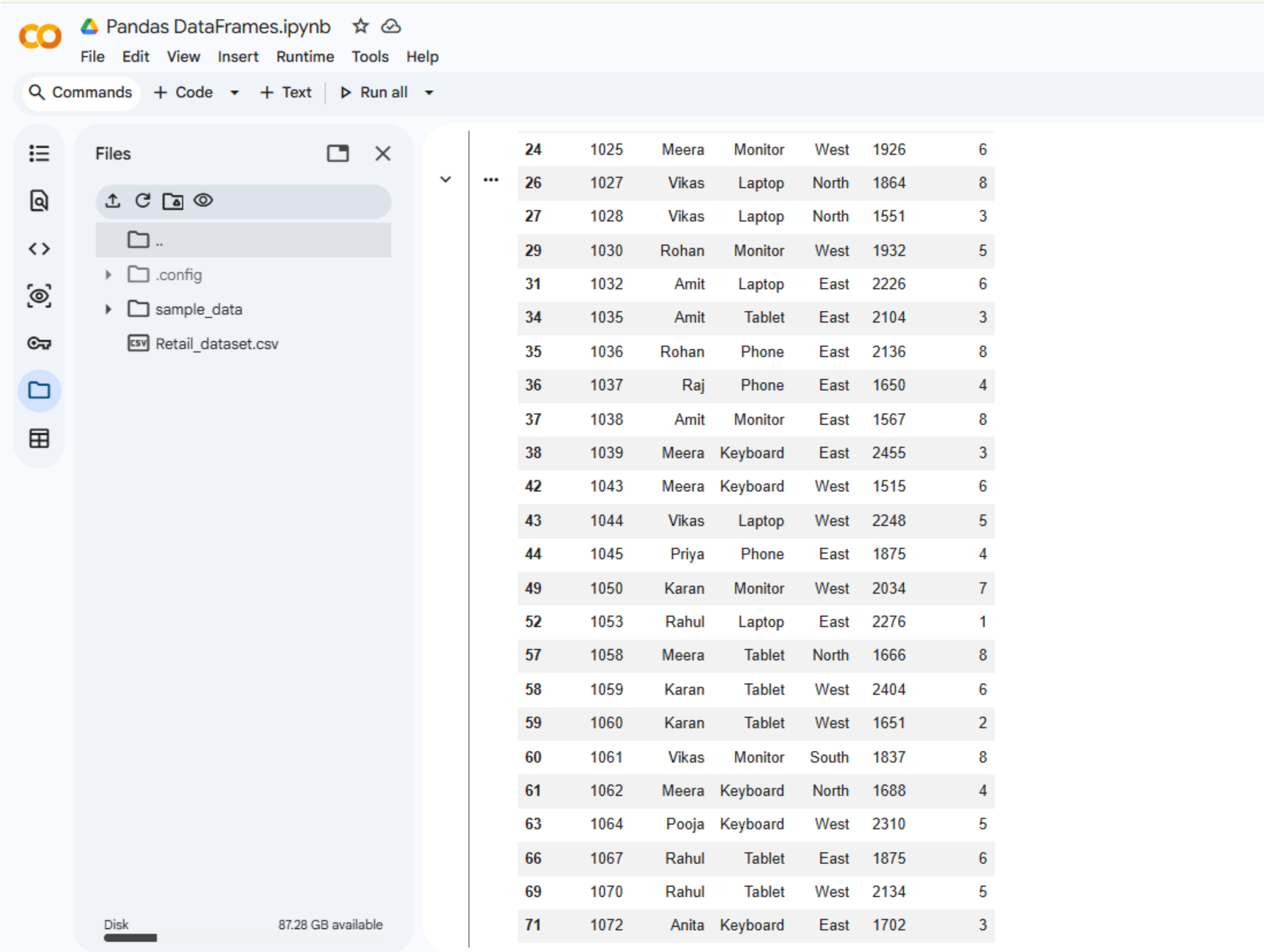
```python
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# 1. Rows where Region = "East"
print("Rows where Region = East:")
display(df[df['Region'] == 'East'])

# 2. Rows where Sales > 1500
print("Rows where Sales > 1500:")
display(df[df['Sales'] > 1500])
```

Rows where Region = East:

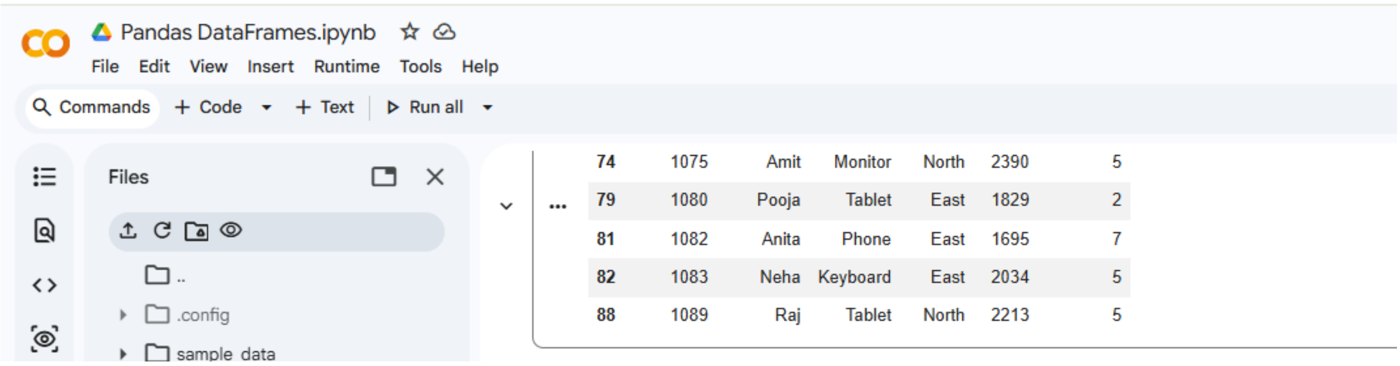
	Order_ID	Customer	Product	Region	Sales	Quantity
7	1008	Sneha	Monitor	East	1269	8
11	1012	Rahul	Laptop	East	1022	5
16	1017	Neha	Laptop	East	2015	2
18	1019	Amit	Phone	East	748	7
23	1024	Karan	Monitor	East	814	7
25	1026	Anita	Tablet	East	839	5
31	1032	Amit	Laptop	East	2226	6
34	1035	Amit	Tablet	East	2104	3
35	1036	Rohan	Phone	East	2136	8
36	1037	Raj	Phone	East	1650	4
37	1038	Amit	Monitor	East	1567	8
38	1039	Meera	Keyboard	East	2455	3
44	1045	Priya	Phone	East	1875	4
47	1048	Rohan	Monitor	East	1103	2
52	1053	Rahul	Laptop	East	2276	1
53	1054	Raj	Monitor	East	1420	2
64	1065	Amit	Monitor	East	815	8
66	1067	Rahul	Tablet	East	1875	6
71	1072	Anita	Keyboard	East	1702	3
76	1077	Vikas	Keyboard	East	978	6
77	1078	Vikas	Laptop	East	538	4
78	1079	Meera	Monitor	East	1295	7
79	1080	Pooja	Tablet	East	1829	2
81	1082	Anita	Phone	East	1695	7
82	1083	Neha	Keyboard	East	2034	5
85	1086	Pooja	Laptop	East	1330	2
87	1088	Karan	Monitor	East	875	2

Rows where Sales > 1500:

	Order_ID	Customer	Product	Region	Sales	Quantity
2	1003	Meera	Laptop	South	1597	5
4	1005	Priya	Phone	South	2028	4
9	1010	Amit	Phone	West	1998	4
10	1011	Raj	Monitor	North	1693	6
13	1014	Rohan	Phone	South	2078	3
16	1017	Neha	Laptop	East	2015	2
17	1018	Anita	Phone	North	1637	3
19	1020	Neha	Laptop	West	1503	6
22	1023	Rahul	Laptop	South	2267	3



24	1025	Meera	Monitor	West	1926	6
26	1027	Vikas	Laptop	North	1864	8
27	1028	Vikas	Laptop	North	1551	3
29	1030	Rohan	Monitor	West	1932	5
31	1032	Amit	Laptop	East	2226	6
34	1035	Amit	Tablet	East	2104	3
35	1036	Rohan	Phone	East	2136	8
36	1037	Raj	Phone	East	1650	4
37	1038	Amit	Monitor	East	1567	8
38	1039	Meera	Keyboard	East	2455	3
42	1043	Meera	Keyboard	West	1515	6
43	1044	Vikas	Laptop	West	2248	5
44	1045	Priya	Phone	East	1875	4
49	1050	Karan	Monitor	West	2034	7
52	1053	Rahul	Laptop	East	2276	1
57	1058	Meera	Tablet	North	1666	8
58	1059	Karan	Tablet	West	2404	6
59	1060	Karan	Tablet	West	1651	2
60	1061	Vikas	Monitor	South	1837	8
61	1062	Meera	Keyboard	North	1688	4
63	1064	Pooja	Keyboard	West	2310	5
66	1067	Rahul	Tablet	East	1875	6
69	1070	Rahul	Tablet	West	2134	5
71	1072	Anita	Keyboard	East	1702	3



74	1075	Amit	Monitor	North	2390	5
79	1080	Pooja	Tablet	East	1829	2
81	1082	Anita	Phone	East	1695	7
82	1083	Neha	Keyboard	East	2034	5
88	1089	Raj	Tablet	North	2213	5

Q8. Multiple Condition Filtering

Display rows where:

1. Region is West
2. Sales is greater than 1200

Ans:-

```python
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# Filter rows where Region is West and Sales > 1200
filtered_data = df[(df['Region'] == 'West') & (df['Sales'] > 1200)]

# Display the result
filtered_data
```

	Order_ID	Customer	Product	Region	Sales	Quantity
9	1010	Amit	Phone	West	1998	4
19	1020	Neha	Laptop	West	1503	6
24	1025	Meera	Monitor	West	1926	6
29	1030	Rohan	Monitor	West	1932	5
42	1043	Meera	Keyboard	West	1515	6
43	1044	Vikas	Laptop	West	2248	5
49	1050	Karan	Monitor	West	2034	7
55	1056	Rahul	Laptop	West	1481	1
58	1059	Karan	Tablet	West	2404	6
59	1060	Karan	Tablet	West	1651	2
63	1064	Pooja	Keyboard	West	2310	5
68	1069	Priya	Laptop	West	1422	6
69	1070	Rahul	Tablet	West	2134	5
73	1074	Meera	Keyboard	West	1429	7
83	1084	Rohan	Laptop	West	1204	5
84	1085	Neha	Monitor	West	1339	7

Q9. Descriptive Statistics

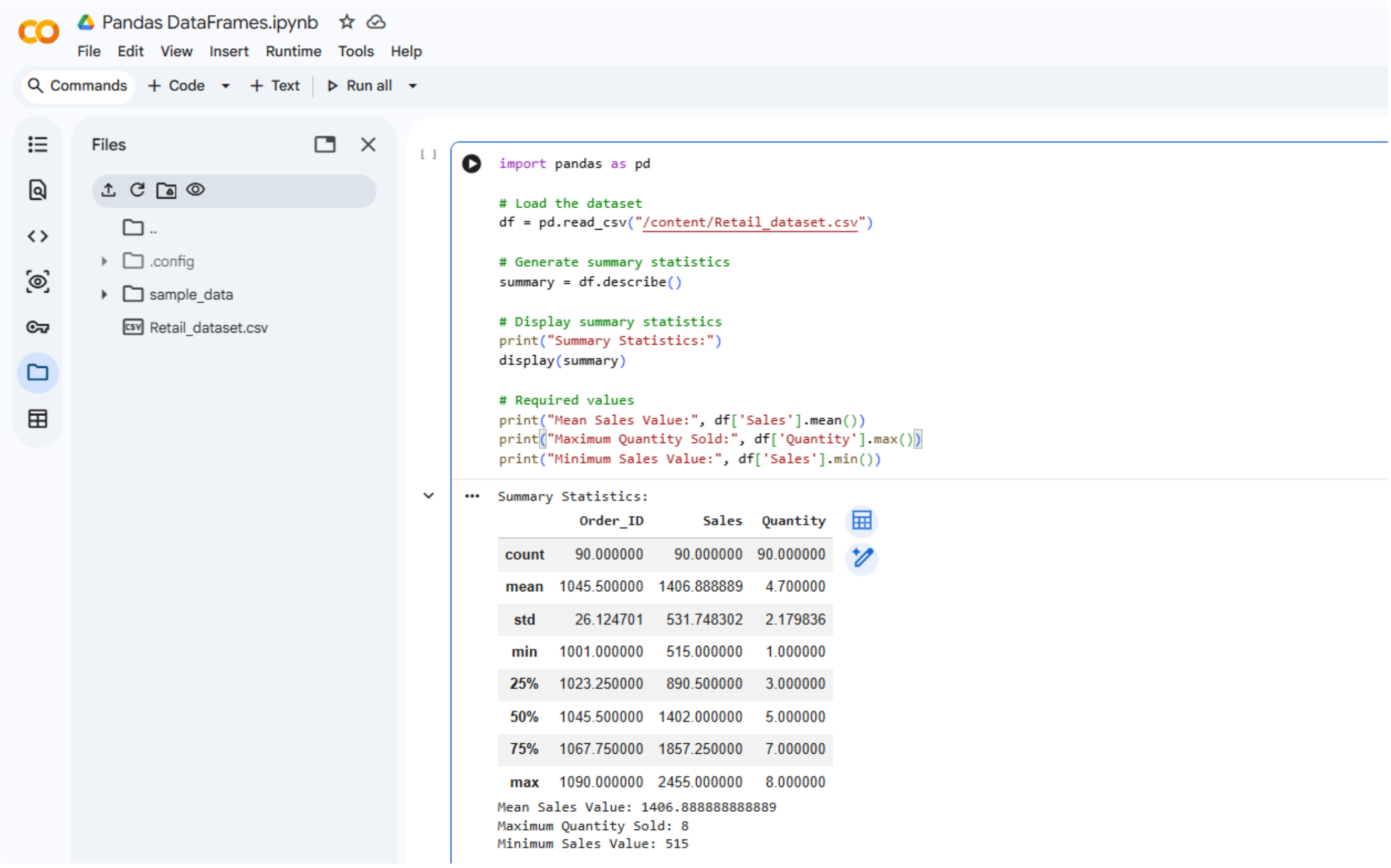
Generate summary statistics using:

describe()

From the result identify:

- Mean sales value
- Maximum quantity sold
- Minimum sales value

Ans:-




```python
import pandas as pd

# Load the dataset
df = pd.read_csv("/content/Retail_dataset.csv")

# Generate summary statistics
summary = df.describe()

# Display summary statistics
print("Summary Statistics:")
display(summary)

# Required values
print("Mean Sales Value:", df['Sales'].mean())
print("Maximum Quantity Sold:", df['Quantity'].max())
print("Minimum Sales Value:", df['Sales'].min())
```

Summary Statistics:

	Order_ID	Sales	Quantity
count	90.000000	90.000000	90.000000
mean	1045.500000	1406.888889	4.700000
std	26.124701	531.748302	2.179836
min	1001.000000	515.000000	1.000000
25%	1023.250000	890.500000	3.000000
50%	1045.500000	1402.000000	5.000000
75%	1067.750000	1857.250000	7.000000
max	1090.000000	2455.000000	8.000000

```
Mean Sales Value: 1406.888888888889
Maximum Quantity Sold: 8
Minimum Sales Value: 515
```

Q10. Aggregation Analysis

Using the Sales column, calculate:

1. Total sales
2. Average sales
3. Maximum sales
4. Minimum sales

Ans:-